

04

LET'S ROLL! IMPLEMENTING THE ATP



Implementation Process Considerations

The development of the project scenarios, cross sections, cost estimates, and implementation strategies translated the MVATP vision and guiding principles into a practical framework for action. This chapter outlines the steps, resources, and decisions needed to change and move projects from concept to construction and policies and programs from ideas to execution, including how to prioritize investments, manage trade-offs, and align projects with available funding opportunities. The implementation approach balances ambition with feasibility, ensuring projects advance safety, comfort, access, sustainability, and biodiversity while working within real-world constraints. Continued public engagement and collaboration with partners are also important components of implementation.

The Green Street scenario demonstrates how street space can be re-envisioned to improve safety, mitigate heat and stormwater issues, and enhance the public realm. By examining material alternatives and identifying ways to manage challenges, it shows why strategic project development matters. Successful implementation relies on collaboration and trust across City departments and with regional partners to advance shared transportation goals.

PRIORITY PROJECT DESIGN CONSIDERATIONS

One of the five Guiding Principles of the MVATP is Action-Oriented and Innovation. To reflect this principle, priority project concepts were developed with two alternatives: resurfacing and reconstruction. Resurfacing represents Action-Oriented, offering lower costs and fewer barriers to implementation. Reconstruction represents Innovation, identifying opportunities to integrate new design guidance, infrastructure concepts, and green street elements.

All the infrastructure recommendations presented in this section will require further public engagement and engineering review to confirm design adequacy and feasibility prior to implementation.

Resurfacing

Resurfacing alternatives assume the street would undergo routine maintenance, including slurry seal or a grind and overlay, creating a “blank slate” for installing bicycle facilities. These alternatives do not involve moving curb lines; therefore, the existing curb-to-curb width defines the space available for reallocation. The scope includes pavement markings, signage, and improved bicycle facilities such as concrete curbs or flexible delineator posts. Sidewalk construction or other concrete work is not included, although minor sidewalk maintenance may be performed to meet Title II ADA requirements. Where projects intersect or cross another agency’s right-of-way (e.g. Caltrans, Santa Clara County), coordination will be needed.

Reconstruction

Reconstruction alternatives assume the street is fully demolished and rebuilt within the existing right-of-way, typically as part of a major capital project triggered by pavement degradation or a similar scale of work. These alternatives allow for curb line adjustments and the installation of continuous Class IV bikeway separation, such as curbed planting strips along the bikeway. No additional right-of-way acquisition is assumed. Reconstruction projects may affect drainage patterns, trees, medians, and other existing features which will require further feasibility analysis. These projects are long-term, visionary concepts that would be adapted to site-specific conditions, recognizing that full implementation may not be practical on narrower streets.

For all alternatives, cross sections were developed based on these assumptions and are detailed in the cutsheets included in Appendix B. Each cutsheet provides a narrative describing existing conditions, proposed changes in each alternative, and implementation considerations, along with an illustrative cross section showing elements and widths. The cutsheets also include maps illustrating project limits and nearby destinations like parks and schools served by the improvements. Cross sections were developed using measurements from high-quality aerial imagery; however, field verification during future design may identify discrepancies that require adjustments.

Costs

Cost estimates for resurfacing alternatives were developed using generalized unit costs (based on city project bid results from 2023 and 2024) and frequency assumptions. These include, but are not limited to, resurfacing, striping, signage, and Class IV bikeway elements. The estimates are intended to reflect the cost of installing bicycle facilities as part of a resurfacing project. Other elements that may be included in resurfacing projects—such as sidewalks or median islands—vary by location and are not included in these estimates.

The estimates also include percentage-based allowances for design, coordination, permitting, environmental review, and similar project development costs, which vary by project scale.

Costs for reconstruction alternatives were not developed. Reconstruction projects typically include substantial additional work unrelated to bicycle or pedestrian facilities, such as utility replacement and full-depth pavement reconstruction, and vary significantly by location. Due to these uncertainties, reliable cost estimates were not feasible. In most cases, the cost of bicycle and pedestrian facilities would represent a relatively small portion of the overall reconstruction project cost.

TABLE 8 ESTIMATED RESURFACING PROJECT COST FOR PRIORITY PROJECTS

PROJECT	LENGTH (MILES)	BIKEWAY CLASS IN RESURFACING ALTERNATIVE	EST. PROJECT COST (2025 DOLLARS IN MILLIONS)
California Street, South Shoreline Boulevard to Showers Drive	1.38	Class IV	\$5.5
California Street, Showers Drive to Del Medio Avenue	0.5	Class IV	\$2.0
Whisman Road, Fairchild Drive to East Dana Street	1.43	Class IV	\$5.7
Escuela Avenue, Crisanto Ave to El Camino Real	0.5	Class III	\$1.5
California Street, Hope Street to South Shoreline Blvd	0.32	Class III	\$1.0
Showers Drive, California Street to El Camino Real	0.32	Class IV	\$1.3
West Middlefield Road, Moffett Blvd to Old Middlefield Way	2	Class IV	\$8.0
Latham Street, Escuela Ave to Showers Drive	0.84	Class III	\$2.5
South Shoreline Boulevard, Montecito Avenue to El Camino Real	1.1	Class IV	\$4.4
Latham St/Church St, Castro Street to Escuela Ave	0.85	Class III	\$2.5
Rengstorff Avenue, Garcia Avenue to Leghorn Street	0.4	Class IV	\$1.6
Dana St, Bush Street Moorpark Way	0.85	Class IV	\$3.4
Miramonte Ave, El Camino Real to Marilyn Drive	0.5	Class IV	\$2.0
Garcia Avenue, Amphitheater Parkway to Bayshore Parkway	0.6	Class II	\$2.9
Evelyn Ave, Bush Street to South Bernardo Avenue	1.35	Class IV	\$5.4
Phyllis Avenue, El Camino Real to Grant Road	0.5	Class IV	\$2.0
Sylvan Ave, Moorpark Way to El Camino Real	0.6	Class IV	\$2.4

Note: Costs are estimates in 2025 dollars, based on information available at the time, and are expected to increase over time. Costs are for resurfacing projects which include unit assumptions for intersections and linear components of a resurfacing project (e.g., markings, signage, temporary materials for protection, pavement resurfacing).

SCENARIO: GREEN STREET INFRASTRUCTURE

Many streets in Mountain View contain excess roadway width that can be reallocated to better support walking, biking, and ecological functions. By shifting curb lines inward where feasible, space can be created along sidewalks to provide separation between people and vehicles, accommodate trees and stormwater features, and improve safety and overall quality of the public realm. While green street infrastructure may involve a greater upfront capital cost, long-term environmental health, traffic safety and social benefits are notable.

FIGURE 32 CONCEPT ILLUSTRATION OF GREEN STREET

Green street infrastructure represents a strategic investment in community well-being, environmental sustainability, and urban livability.



Material Alternatives

Based on block-level analysis, six material alternatives were evaluated against a base street modification cost of \$2.82 million (\$535 per foot of roadway) to create a 4-foot green infrastructure zone. Base costs assume one mile of street reconstruction and each material alternative's cost would be added to that. The estimate includes both sides of the street with no change to the overall roadway width. These estimates are intended to indicate relative costs rather than final construction prices.

The estimate assumes full reconstruction of sidewalks and driveways within the right-of-way. Driveways are assumed to occur every 100 feet and to be approximately 20 feet wide. Cost data are based on aggregated City of Mountain View bid prices for similar projects and were rounded. Figure 33 illustrates the costs of various material alternatives for the reimagined street section.

FIGURE 33 ADD-ON COST PER MILE FOR MATERIAL ALTERNATIVES IN REIMAGINED STREETS BEYOND BASE STREET MODIFICATION COST



Benefits

- ▶ Mulch and aggregate provide immediate cost advantages over concrete.
- ▶ Drought-tolerant planted options begin providing environmental benefits early, such as temperature reduction and improved pedestrian comfort.
- ▶ Concrete provides durability, low maintenance, and predictable performance.
- ▶ Planted alternatives with irrigation and bioswales deliver growing returns over time such as expanded tree canopy, better air quality, stabilized maintenance costs, and cost-effective compliance with the Municipal Regional Stormwater Permit (MRP 3.0).

Trade-Offs

- ▶ Concrete has high upfront costs and limited ecological or community benefits.
- ▶ Mulch and aggregate may require more frequent replenishment or upkeep, adding to ongoing maintenance costs.
- ▶ Planted options need irrigation and care during establishment.
- ▶ Irrigated and bioswale-planted landscapes require higher initial investment. These plantings will also require maintenance such as irrigation and pruning after maturity.

Challenges and Mitigations

Moving curb lines can significantly increase project complexity and cost by triggering a range of additional infrastructure requirements. These include, but are not limited to: drainage analysis and utility adjustments; sidewalk, signal, and driveway demolition and reconstruction; and extensive coordination with property owners. It requires more complex engineering, construction, and interagency coordination beyond the basic street maintenance. Table 9 outlines the key considerations associated with green street projects and practical ways to reduce or manage them.

TABLE 9 COMMON GREEN STREET CHALLENGES AND POTENTIAL SOLUTIONS

CHALLENGE	MITIGATION
Higher upfront capital costs	Upfront costs can be reduced or offset by incorporating into capital projects early in a project life cycle, phasing improvements, and pursuing grant funding, while the environmental, aesthetic, and infrastructure benefits persist for the life of the tree and associated improvements.
Tree roots can lift sidewalks	Careful species selection, appropriate planting techniques, and engineered solutions such as structural soils and modified sidewalk sections can provide adequate root space and reduce the risk of root-related damage.
Potential property damage from falling limbs	Tree species with strong branch attachment, appropriate placement, and routine maintenance reduce limb risk without limiting tree size. Small or medium varieties are selected only in response to valid concerns from adjacent property owners.
Reduced visibility within the right-of-way	Tree placement will be carefully coordinated to avoid sight triangles and maintain safe visibility at intersections and driveways.
Requires ongoing maintenance	All maintenance will be performed by the City as part of its standard operations.

Implications of Green Streets

Green streets deliver measurable benefits that extend far beyond aesthetics. They provide space for natural elements to support biodiversity, incorporate sustainable stormwater strategies, create inviting environments through complementing traffic-calming efforts and prioritize active transportation modes for people of all ages and abilities. Simply put,

Green Streets = Active Transportation + Biodiversity + Green Infrastructure.

“I think trees can do wonders in making some of the overwhelming walking areas (e.g. along Central around Rengstorff) less so and make it more pleasant and psychologically feel safer.”

Figure 34 represents a range of green street elements and its benefits. Green stormwater infrastructure is required for Municipal Regional Stormwater Permit 3.0 compliance, managing runoff through increased pervious surfaces that reduce, slow, and treat stormwater while improving water quality. Street trees and vegetation enhance air quality by filtering pollutants and, when using native species, provide habitat corridors that support pollinators and ecosystems in this biodiversity hotspot region.

GREEN STREETS

Mountain View’s tree cover is 19.5% of the city, with a goal of 22.7% by 2030.

Refer to the Biodiversity and Urban Forest Plan for more information about green streets progress, goals, and actions.

FIGURE 34 A FEW GREEN STREET ELEMENTS AND THEIR BENEFITS

Planting strip: Replaces heat-absorbing pavement; absorbs rainwater and reduces runoff



Cul-de-sac: Captures stormwater and improves local drainage



Tree well: Supports tree growth, providing cooling streets and shade for pedestrians



Bikeway: Provides safe, comfortable, and shaded routes



Parklet: Creates public spaces for rest and social interaction



Median: Filters air pollution and adds greenery to busy roads

Green street projects require strategic site selection, interagency coordination, and supporting policies. Engaging the residents and businesses early helps build support, address concerns about visibility and maintenance, and ensure designs reflect community priorities. Implementation strategies that reduce risk and build momentum include:

- ▶ **Pilot and phased projects:** Start with lower-risk streets that have favorable conditions to demonstrate feasibility. Use a phased approach that allows vegetation to establish while monitoring safety, performance, and maintenance needs. Lessons learned from these pilots will inform expansion to more complex or higher-priority locations in future years.
- ▶ **Target priority corridors and nodes:** Once pilot projects demonstrate success, focus investments on high-priority routes such as streets near schools, commercial areas, flood-prone segments, and existing or planned bikeways and trail connections. This approach advances both stormwater compliance and active transportation goals simultaneously.
- ▶ **Standardize designs:** Use simple, repeatable typologies such as corner curb extensions with bioretention, mid-block rain gardens, and permeable parking lanes to streamline design, permitting, construction, and maintenance.

Good candidates for these treatments include streets that are in need of utility upgrades or repairs, have excess pavement width, have drainage conditions that allow for curb line modifications, and supportive property owners. Any proposed green areas should be planned with consideration of existing underground utilities to avoid conflicts and ensure feasibility during installation and maintenance. Identifying streets with these characteristics can substantially reduce risk, cost, and implementation challenges.

Funding Sources for Active Transportation Projects

Funding to support active transportation infrastructure and programming is available from many different sources at the federal, state, regional and local level. Mountain View often uses local funds to support transportation planning studies, design, and environmental review stages of transportation infrastructure projects, and utilizes regional, state, and federal grant funds for construction. Programs such as Safe Routes to School are also funded with municipal funds.

A non-exhaustive list of potential funding options includes (Table 10):

Federal Funding

Safe Streets and Roads for All (SS4A)

SS4A is a federal-aid grant program for regional, local, and Tribal initiatives to improve roadway safety. Funding is provided for Planning and Demonstration Grants that support the development of a comprehensive safety action plan that identify the most pressing roadway safety concerns in a community and implement solutions. Implementation Grants fund projects and initiatives to address identified roadway safety issues. An eligible Action Plan must be adopted before applying for implementation funds.

Highway Safety Improvement Program (HSIP)

HSIP is a federal-aid program that supports states' efforts to significantly reduce fatal and serious injuries on public roads with spot or systemic safety improvements. California splits HSIP funds between state HSIP for state highways and local HSIP for local roads. Recipient agencies are required to have a completed Local Road Safety Plan (LRSP) or an equivalent plan to be eligible.

Community Development Block Grant (CDBG)

The CDBG program is a federal grant program funded by the Department of Housing and Urban Development (HUD) to support various projects and programs serving low-income areas. CDBG funds may be used for transportation infrastructure projects such as sidewalk construction, provided they expand economic opportunities for low- and moderate-income households through increased mobility, safety, and accessibility. Mountain View receives about \$600,000 each year in CDBG funds that are primarily allocated to non-profit agencies on a two-year application cycle.

State/Regional Funding

California Transportation Commission Active Transportation Program (ATP)

The Active Transportation Program consolidated various transportation programs into a single program and was originally funded at about \$123 million a year from a combination of state and federal funds. The goals of the ATP include increasing trips completed by walking and biking, increasing the safety and mobility of pedestrians, advancing climate and public health goals, and supporting projects that benefit many types of users including disadvantaged communities. Other Santa Clara County cities have previously been awarded ATP funds to fund pedestrian and bicycle safety projects.

Strategic Growth Council Affordable Housing and Sustainable Transportation Program (AHSC)

As a grant program of the State of California, the AHSC Program funds affordable housing and transportation projects close to jobs, schools, and other daily destinations to help California meet both its climate and equity goals. AHSC provides funding for affordable housing developments, sustainable transportation infrastructure, and programs that encourage residents to walk, bicycle, and use public transit. Mountain View received \$49.7 million to transform an underutilized parking lot into a 7-story development with 161 apartment units within a ten-minute walk of a major multimodal transit station with bus, light rail, and commuter rail services.

Transportation Development Act Article 3 (TDA 3)

The TDA was enacted by the California Legislature in 1971 to improve existing public transportation services and encourage regional transportation coordination. It allocates tax revenue funding for transit and non-transit related purposes that comply with regional transportation plans. Eligible counties with a population under 500,000 may also use funds for local streets and roads, construction and maintenance. TDA 3 funds (one-quarter cent of the general statewide sales tax) are a subset of the overall TDA funds collected specifically for bicycle and pedestrian projects. The Metropolitan Transportation Commission (MTC), the nine-county MPO in the San Francisco Bay Area, allows counties to use these funds for projects that are reviewed by a Bicycle Advisory Committee and approved by MTC. While the amount of annual TDA3 funding allocated varies and depends upon the total sales tax revenues generated, Mountain View in past years has been allocated approximately \$120,000 annually from this funding source. The City has used past TDA3 grant funds for a pedestrian and bicycle infrastructure improvements and program and policy elements such as city bike maps, bike racks, and education programs. Mountain View coordinates with VTA to access these funds and can bank up to five years of funds if needed to plan for larger projects.

Santa Clara Valley Transportation Authority (VTA) 2016 Measure B

Santa Clara County voters approved 2016 Measure B, a 30-year, half-cent countywide sales tax to enhance transit, highways, expressways, and active transportation. Collection began in April 2017. Funds are allocated across nine Program Categories, including Local Streets and Roads, BART Phase II, Caltrain Grade Separations, Transit Operations, and Bicycle and Pedestrian. The Bicycle and Pedestrian category funds projects of countywide significance, with priority given to projects that connect to schools, transit, and employment centers, fill gaps in the existing bicycle and pedestrian network, safely cross barriers to mobility, and make bicycling and walking more convenient for county residents and visitors.

Bay Area Air Quality Management District (BAAQMD) Transportation for Clean Air (TFCA) Regional Fund

Since 1991, the Department of Motor Vehicles has imposed a \$4 surcharge on cars and trucks registering within the BAAQMD, also known as the “Air District”, to provide grant funding to projects that reduce motor vehicle emissions. The Air District’s Board of Directors approves Calls for Projects and allocates 60% of TFCA funds every year. The other 40% of funds are passed through nine Bay Area counties through the TFCA 40 Percent Fund.

Local Funding

California Transportation Commission (CTC) Local Partnership Program (LPP)

Using an annual \$200 million appropriation from the Road Maintenance and Rehabilitation Account, the LPP provides funding via both a formulaic and competitive process to counties, cities, districts, and regional transportation agencies that have approved fees or taxes dedicated to transportation improvements. LPP funds are used to improve aging infrastructure, road conditions, active transportation, transit and rail, and health and safety benefits. While Mountain View is not directly eligible for these funds, they may coordinate with regional agencies such as but not limited to VTA, Bay Area Toll Authority (BATA), and Caltrain to submit projects for consideration.

TABLE 10 OVERVIEW OF POTENTIAL FUNDING OPTIONS

FUNDING MECHANISM	ALIGNMENT	DESCRIPTION
Federal		
Safe Streets and Roads for All (SS4A)	Policy	Federal grant program for initiatives to improve roadway safety projects that reduce or eliminate roadway fatalities
Highway Safety Improvement Program (HSIP)	Policy	Federal-aid program supporting states' efforts to significantly reduce fatal and serious injuries on public roads.
Community Development Block Grant (CDBG)	Policy	Federal grant program by HUD to support various projects and programs serving low-income areas, including infrastructure projects that expand economic opportunities.
State/Regional		
California Transportation Commission Active Transportation Program (ATP)	Policy	State program disbursing funds to active transportation projects that promote walking and biking trips, pedestrian safety and mobility, and climate and public health goals.
Strategic Growth Council Affordable Housing and Sustainable Transportation Program (AHSC)	Policy	State grant program funding affordable housing and transportation projects close to jobs, schools, and other daily destinations that encourage residents to walk, bicycle, and use public transit.
Santa Clara Valley Transportation Authority (VTA) 2016 Measure B	Fiscal	County tax measure enacting a 30-year, half-cent sales tax to enhance transit, highways, expressways, and active transportation. Funds are provided for nine program categories, including Bicycle and Pedestrian.
BAAQMD Transportation for Clean Air (TFCA)	Fiscal	\$4 surcharge on cars and trucks registering within the Bay Area Air District. Funds are used for grants to projects that reduce motor vehicle emissions.
Transportation Development Act Article 3 (TDA3)	Fiscal	Section of the 1971 TDA that specifically funds biking and pedestrian projects. Eligible counties with a population under 500,000 may also use funds for local streets and roads, construction and maintenance.
Local		
CTC Local Partnership Program (LPP)	Fiscal	An annual \$200 million appropriation for funding counties, cities, districts, and regional transportation agencies that have approved fees or taxes dedicated to transportation improvements. Funds can be used to support initiatives to improve infrastructure, mobility, and public health.

COLLABORATION

Coordination and collaboration are critical for the implementation of projects, programs and policies. The complexity of coordination was a factor when scoring projects, considering not just internal collaboration among City departments but also the level of coordination needed with regional and on any given project with local, regional, and state partners such as Caltrans, VTA, Santa Clara County, and Mountain View's neighboring cities of Sunnyvale, Los Altos, and Palo Alto to move projects toward implementation. Additional partners adds complexity to projects and can extend the overall project timeline due to the increased amount of coordination and engagement, but is necessary to achieve shared transportation goals, particularly for corridors that span multiple jurisdictions. A recent example of effective cross-agency collaboration is the El Camino Real protected bikeways, installed in partnership with Caltrans to improve multimodal access along this key corridor. State level agency focus in recent years on Complete Streets, state highways as Main Streets, vulnerable road users and transit priority suggest more alignment and coordination on local and state level priorities going forward.

PUBLIC INPUT

Active transportation is a part of everyone's life in Mountain View. The City is strongly committed to public involvement and will continue to engage as the MVATP is implemented. Public input allows the city to understand how a street is being used and learn concerns that may not be obvious from quantitative data sources.

For smaller bicycle and pedestrian projects, the City typically seeks public input at early design stage, through public presentations at the Bicycle/Pedestrian Advisory Committee (BPAC), Council Transportation Committee (CTC), and approval of design at City Council. After design is approved, the project goes out to bid and construction begins. During construction, the city notifies nearby property owners of the construction.

For larger projects or projects that significantly modify the transportation system, the City may conduct a feasibility study to identify a preferred concept before beginning engineering design. Feasibility studies are an extended planning process to understand existing conditions, establish goals for the project, develop several alternative concepts, analyze how the concepts support project goals and impact transportation safety and operations, seek community input on the concepts, and bring a preferred concept to City Council for approval. The BPAC and CTC are typically updated and invited to provide input at key points during the feasibility study. Outreach strategies the City has used for feasibility studies include walking or biking tours of the project area, public workshops, online surveys, presentations to local groups and organizations, mailers to adjacent properties, and other ways of soliciting public input. The City ensures materials are multi-lingual and offers interpretation and translation services.

The City will seek public input as it advances the priority projects in the MVATP. The level of engagement will generally align with the size or impact of the project, as described above. Resurfacing projects that do not significantly alter the transportation system will typically follow the small project process. Reconstruction projects and those that significantly alter the transportation system will go through additional analysis, up to and including a feasibility study, before engineering design begins.

PRIVATE DEVELOPMENT AND THE MVATP

Mountain View is committed to expanding and growing in a way that supports housing and active transportation. Through the development review process, the City works with developers to ensure they are constructing the planned improvements identified in adopted city plans, such as Precise Plans, Utility Master Plans and this ATP. This plan recognizes the need to strengthen tools the City uses to ensure developments support the City's active transportation vision.

COMMITMENT TO THE VISION AND NEXT STEPS FOR THE CITY

Active transportation is an integral part of life in Mountain View, as evidenced by the high rates of walking and bicycling by those who travel about the city, and by the city and community's investment in active transportation programs and facilities.

Using the Guiding Principles of Mobility & Connectivity, Safety & Comfort, Access & Equity, Sustainability & Biodiversity, and Innovation & Action-Oriented, the MVATP identified sustainable projects that improve safe, equitable and comfortable connections across the city and programs and policies that continue to support active transportation. This list of projects, programs and policies provides an action plan for the next 5-10 years for the city to move projects forward in the capital planning process. The city's commitment to multimodal transportation mobility, safety, and connectivity can be measured through the progress and timely delivery of the priorities outlined in this plan.

The MVATP is a commitment by the city to creating an active transportation system that strengthens the community's access to housing, employment, parks, schools and other destinations and enhances biodiversity. The plan's priority projects, programs and policies complement other Mountain View planning documents and initiatives that support active transportation, safety, sustainability and biodiversity and will help to provide accountability to the community's commitment to these values. The MVATP will also help the City continue to be a leader in the region, and the state, in community walking and bicycling activity.



City of
Mountain View